

$$3.2 \quad \langle f, g \rangle = \int_1^5 f(x)g(x) dx$$

a.) $LO\{1, x, x^2\}$ - ^{v_1, v_2, v_3} Pronalezení ON báze provedeme algoritmus G-S. ortogonalizace

$$u_1 = \frac{v_1}{\|v_1\|}$$

$$\vec{u}_1 = \frac{1}{\sqrt{5}} \cdot 1 = \underline{\underline{\frac{1}{2}}}$$

w_2 - O. projekce v_2 na podprostor $LO\{\frac{1}{2}\}$

$$w_2 = \langle u_1, v_2 \rangle \cdot u_1 = \int_1^5 \frac{1}{2} x dx \cdot \frac{1}{2} = 6 \cdot \frac{1}{2} = 3$$

$$u_2 = \frac{v_2 - w_2}{\|v_2 - w_2\|} = \frac{x - 3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \sqrt{\frac{3}{16}} \cdot (x-3) = \underline{\underline{\frac{\sqrt{3}}{4} \cdot (x-3)}}$$

$$w_3 = \langle u_1, v_3 \rangle u_1 + \langle u_2, v_3 \rangle u_2 = \int_1^5 \frac{x^2}{2} dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3} \cdot (x-3) \cdot x^2}{4} dx \cdot \frac{\sqrt{3}}{4} \cdot (x-3) = \frac{31}{3} + 6 \cdot (x-3)$$

$$v_3 - w_3 = x^2 - \frac{31}{3} - 6x + 18 = x^2 - 6x + \frac{23}{3}$$

$$u_3 = \frac{v_3 - w_3}{\|v_3 - w_3\|} = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \sqrt{\frac{45}{256}} \cdot (x^2 - 6x + \frac{23}{3}) = \underline{\underline{\frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3})}}$$

ON báze: $B = \left(\frac{1}{2}; \frac{\sqrt{3}}{4} \cdot (x-3); \frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3}) \right)$

b.) w - ort. projekce funkce $\sin(x)$ na $LO\{1, x, x^2\}$... využijí ON bázi;

$$w = \langle u_1, v \rangle u_1 + \langle u_2, v \rangle u_2 + \langle u_3, v \rangle u_3 = \int_1^5 \frac{\sin(x)}{2} dx \cdot \frac{1}{2} + \int_1^5 \frac{\sqrt{3} \cdot (x-3) \cdot \sin(x)}{4} dx \cdot \frac{\sqrt{3}}{4} \cdot (x-3) + \int_1^5 \frac{3\sqrt{5} \cdot (x^2 - 6x + \frac{23}{3}) \cdot \sin(x)}{16} dx \cdot \frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3}) = \frac{1}{2} \cdot (\cos(1) - \cos(5)) \cdot \frac{1}{2} + \frac{\sqrt{3}}{4} \cdot (\sin(5) - 2\cos(5) - \sin(1) - 2\cos(1)) \cdot \frac{\sqrt{3}}{4} \cdot (x-3) + \frac{\sqrt{5}}{16} (12\sin(5) - 2\cos(5) + 12\sin(1) + 2\cos(1)) \cdot \frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3})$$

- Po dopočítání vychází následující koeficienty:

$$w = \underline{\underline{0,088x^2 - 0,879x + 1,518}}$$

$$c.) \| \sin(x) - (ax^2 + bx + c) \| = \sqrt{\langle \sin(x) - (ax^2 + bx + c), \sin(x) - (ax^2 + bx + c) \rangle} = \sqrt{\int_1^5 \sin^2 x - 2\sin x \cdot (ax^2 + bx + c) + (ax^2 + bx + c)^2 dx}$$

$$\underline{\underline{= 0,318}} \quad \dots \text{po dosazení koeficientů } a, b, c$$